Transfusion clinique et biologique 30 (2023) 272–278

Available online at ScienceDirect www.sciencedirect.com

Elsevier Masson France EM|consulte www.em-consulte.com/en

Original article

Hepatitis C virus among blood donors in Lubumbashi, DRC: Seroprevalence and molecular characterisation

A. Kabamba-Tshikongo [a,b,e,*], H. Manya-Mboni [a], C. Mwamba-Mulumba [c], N.B. Takaisi-Kikuni [d], A.T. Vandenbroucke [e], A.T. Pâques [e], G. Dessilly [e], B. Kabamba-Mukadi [e], A. Longanga-Otshudi [a,b]

[a] Laboratoire de Biologie Clinique, Faculté des Sciences Pharmaceutiques, Université de Lubumbashi, Lubumbashi, Democratic Republic of the Congo
[b] Centre d'Excellence et d'Expertise des hépatites virales et autres pathologies, Lubumbashi, Democratic Republic of the Congo
[c] Faculté de Médecine, Université de Lubumbashi, Lubumbashi, Democratic Republic of the Congo
[d] Laboratoire de Microbiologie Expérimentale et Pharmaceutique, Faculté des Sciences Pharmaceutiques, Université de Kinshasa, Kinshasa, Democratic Republic of the Congo
[e] Institut de Recherche Expérimentale et Clinique, Pôle de Microbiologie, Université Catholique de Louvain, Brussels, Belgium

ARTICLE INFO

Article history:
Available online 8 March 2023

Keywords:
Seroprevalence
Genotypes
HCV
Resistance
Blood donors
DRC

ABSTRACT

Objectives: To date, no study has been done yet on the distribution of Hepatitis C virus genotypes in Lubumbashi, Democratic Republic of Congo. The objective of this work was to determine the seroprevalence and study the distribution of hepatitis C virus (HCV) genotypes among blood donors in Lubumbashi, DRC.
Methods: This was a cross-sectional descriptive study among blood donors. The detection of anti-HCV antibodies was carried out by rapid diagnostic test (RDT) then confirmed by Chemiluminescent immuno-assay (CLIA). Viral load was determined by Nucleic Acid Amplification test (NAT) on Panther system and genotyping by Next Generation Sequencing (NGS) on Sentosa platform.
Results: The obtained seroprevalence was 4.8%. Genotypes 3a (5.0%), 4 (90.0%) and 7 (5.0%) and a few drug resistance mutations were identified in the study population. Significant disturbances of some studied biochemical parameters (HDL-cholesterol, direct bilirubin, transaminases, ALP, GGT and albumin) have been observed in positive HCV blood donors. Irregular family and volunteer donors have been found as the socio-demographic characteristics associated with hepatitis C.
Conclusion: With a seroprevalence of 4.8% obtained among blood donors, Lubumbashi is in an area with medium endemicity for HCV, highlighting the need to implement strategies aiming to improve transfusion safety among blood recipients in Lubumbashi. This study reports for the first time the presence of HCV strains of genotypes 3a, 4 and 7. These results might allow better therapeutic management of HCV infections and contribute to the development of the mapping of HCV genotypes in Lubumbashi and DRC as well.
© 2023 Société française de transfusion sanguine (SFTS). Published by Elsevier Masson SAS. All rights reserved.

1. Introduction

Over the past twenty years, safety in blood transfusion has made great progress regarding the infectious and immunological risk. According to the Congolese blood transfusion policy, screening for human immunodeficiency virus (HIV), hepatitis B (HBV) and C (HCV) virus, syphilis and malaria are among the mandatory examinations on the blood being donated [1,2].

Chronic HCV infection is a public health problem, affecting more than 71 million people and causing approximately 500,000 deaths each year. It is estimated that in 2015 there were 1.75 million new cases of viral hepatitis C (VHC). Only 20% of people with chronic hepatitis C were aware of their infection. In developed countries, advances in screening for hepatitis C have reduced considerably the risk of contamination by blood transfusion. However, in developing countries, blood transfusion remains one of the most common risk factors for developing HCV infection [3,4,5]. Majority of blood donors are family donors and efficient laboratory screening is still a big challenge in many African countries, such as the DRC [6].

HCV is an enveloped positive single-stranded RNA virus, which belongs to the *Flaviviridae* family, genus hepacivirus. HCV is

* Corresponding author at: Laboratoire de Biologie Clinique, Faculté des Sciences Pharmaceutiques, Université de Lubumbashi, Lubumbashi, Democratic Republic of the Congo.
E-mail addresses: arsene.kabamba@gmail.com, tshikongok@unilu.ac.cd (A. Kabamba-Tshikongo).

https://doi.org/10.1016/j.tracli.2023.02.007
1246-7820/© 2023 Société française de transfusion sanguine (SFTS). Published by Elsevier Masson SAS. All rights reserved.

characterized by high genetic variability. The phylogenetic analysis of HCV strains isolated in different regions of the world has identified eight genotypes, numbered from 1 to 8 [7]. There is also a great diversity within the genotypes in the countries of central and western Africa [8,9]. These genotypes may display different biological properties, influence the course of liver disease and the response to treatment [10]. The control of viral hepatitis C has so far been able to make very little progress. However, in recent years, therapeutic advances based on HCV genotyping have significantly improved the management of patients with better responses to treatment, even if there is still a significant proportion of patients who are not-responders to treatment [11].

With a seroprevalence rate estimated at 3%, sub-Saharan Africa might be the World Health Organization (WHO) region with the highest HCV prevalence. With regard to the Democratic Republic of Congo (DRC), apart from scattered studies [5,8], epidemiological data at the national level are not reported.

HCV screening in blood banks and reference laboratories is carried out by detecting antibodies (Ab) against the virus and/or the presence of viral RNA in serum or plasma [2,4]. Rapid diagnostic tests (RDTs) remain important tools for blood donor screening in the DRC for these infections. However, techniques for inactivating pathogens possibly present in labile blood derivatives are not currently applicable in Lubumbashi and whole blood is generally used during transfusions instead of concentrated red blood cells. Epidemiological studies of these infections in blood donors allow to monitor the prevalence, identify risk factors and the main means of combating and preventing their dissemination through transfusion. The aim of this work was to determine the seroprevalence of hepatitis C viral markers in blood donors and study the distribution of RNA-HCV genotypes among the blood donor population in Lubumbashi, Democratic Republic of Congo.

2. Materials and methods

2.1. Site, population, type, and study period

This was a cross-sectional descriptive study of the prevalence of anti-HCV antibodies in blood donors. The site, population, study period and parameters have been previously described [12]. Briefly, the study took place from November 1, 2017 to December 30, 2019 in two hospitals of Lubumbashi: Hôpital Jason Sendwe and Cliniques Universitaires de Lubumbashi. These two hospitals were selected because they have functional blood banks that serve the population of Lubumbashi and its surroundings. The study population included women and men aged from 18 to 65, in apparent good health, with a hemoglobin level $\geq$ 12 g/dL. In Lubumbashi, the blood donor selection process consists of three stages: pre-screening; clinical and laboratory screening. Pre-screening consists of the measurement of pulse, body mass, blood pressure, hematocrit or hemoglobin. Clinical screening is conducted by interview using a questionnaire related to the candidate's risk of inability to donate. Risk factors related to HCV infection included: tattooing, piercing, surgery in the last 6 months, tooth extraction, sexually transmitted diseases, risky behavior (needle sharing, prostitution), drug abuse and transfusion history. Other factors included the frequency of donations, minimum and maximum age allowed, pregnancy, lactation, menstruation and the normalization of the parameters measured during the pre-screening phase. Once the blood donor candidate satisfies the conditions of the clinical screening phase, he proceeds to the 3rd and last phase where the blood donation and the blood sample are taken to be sent to the biomedical analysis laboratory.

The sample size was 1512. It was determined using the single population proportion formula, based on the previous prevalence of positive HCV antibodies (Ab) in blood donors in Lubumbashi (DRC) estimated at 10% [13].

2.2. Screening for hepatitis C virus

The "One Step Hepatitis C Virus Test Strip (Accurate, China)" RDT was used as screening test. Reactive samples for anti-HCV were re-examined using a confirmation test, the CLIA (Liaison®XL murex HCV Ab assay on the Liaison®XL analyzer, Diasorin, Italy). Only reactive samples in RDT and confirmed in CLIA (Liaison®XL) were considered as positive samples and used in the determination of seroprevalence. The screening and confirmatory tests have been certified by WHO for "One step Hepatitis C Virus Test Strip" and CE marked for "Liaison XL".

2.3. Biochemical evaluation

The evaluation of biochemical parameters in blood donors involved a group of 33 randomly selected anti-HCV positive blood donors. The evaluated biochemical parameters have been previously described [12]. These parameters were measured by the spectrophotometric method on the Cyan Smart machine (Cypress Diagnostics, Belgium).

2.4. Molecular characterization of the hepatitis C virus

2.4.1. HCV viral determination

HCV viral load was quantified on the Panther® system. The Aptima HCV Quant Dx Assay is a nucleic acid amplification test that uses transcription-mediated amplification (TMA) to detect and quantify HCV RNA. This test targets a conserved region of the HCV genome for the detection and quantification of different genotypes. The evaluated detection limit of HCV-RNA in NAT (TMA) was 10 IU/mL, and the specificity of 100% with a linearity test ranging from 10 to10^8 IU/mL of viral load.

2.4.2. RNA-HCV- sequencing

Sequencing was performed using the Sentosa® SQ HCV genotyping test (Vela Diagnostics), a next-generation sequencing-based *in vitro* diagnostic test for viral genotyping of the hepatitis C virus from human plasma or serum.

2.4.3. HCV genotyping and drug resistance

After sequencing, we aligned the obtained sequences using the "Geneious version 11.1″ computer program (Biomatters, Auckland, New Zealand). We identified the genotype by performing a "Nucleotide Blast" of the consensus sequence using National Center for Biotechnology Information (NCBI) software. HCV antiviral resistance mutations were identified, and genotypes were confirmed by the online software geno2pheno (Max Planck Informatic Institute, Germany).

Phylogenetic analysis of HCV genotypes was determined by the alignment of sequences from the present study to HCV reference sequences recovered from GenBank. The phylogenetic tree was built based on the neighborhood method with the Geneious software, according to the Tamoura-Nei model. The sequencing analysis allowed genotyping with a minimum HCV- RNA viral load of 500 IU/mL.

2.5. Data processing and analysis

The collected data were encoded, processed, and analyzed using Microsoft Excel 2010 software and EPI INFO™ version 7.2.4.0 (USA). Mean and standard deviation were calculated and Chi-square test or, if necessary, Fischer's exact test was used to find the association between socio-demographic characteristics and anti-HCV

positivity, and other analyzed parameters. Differences were considered statistically significant at p-value < 0.05.

2.6. Ethical approval

Free and informed consent was obtained from each participant. This study had obtained the approval of the medical ethics committee of the University of Lubumbashi, under the number: UNILU/CEM/095/2017.

3. Results

3.1. Sociodemographic characteristics and HCV seroprevalence

During the study period, 1512 blood donors were included. The seroprevalence of anti-HCV antibody in blood donors was 4.8% (73 out of 1512 tested blood donors) (Fig. 1). Sociodemographic characteristics such as male gender (OR = 2.25), blood donors in the 30–44 age group (OR = 2.6) and family donors (OR = 3.9) were correlated to this seroprevalence (Table 1).

3.2. Biochemical evaluation

Comparison was made between HCV positive blood donors and healthy blood donors according to their biochemical parameters. The results of this evaluation are presented in Table 2.

3.3. Molecular characterisation

Of the 73 HCV positive blood donors, 48 (3.2%) were found to be reactive in NAT (TMA). However, only 20 of them having a viral load > 500 IU/mL were selected for sequencing. Genotypes 3a (5.0%), 4 (90.0%) and 7 (5.0%) were identified among the 20 studied HCV sequences. The identified genotype 4 strains belonged to the 4c (50.0%), 4 k (33.3%) and 4r (16.7%) subgenotypes (Table 3). The alignments of different nucleotide sequences of HCV with respect to the reference sequences confirmed these genotypes (Fig. 2). The phylogenetic analysis of all these HCV strains showed that the identified HCV genotype 3 was close to that found in China, and the subgenotypes of HCV genotype 4 were mostly close to those found in Cameroon, while the identified HCV of genotype 7 was close to those isolated in Uganda from two congolese citizen from the DRC.

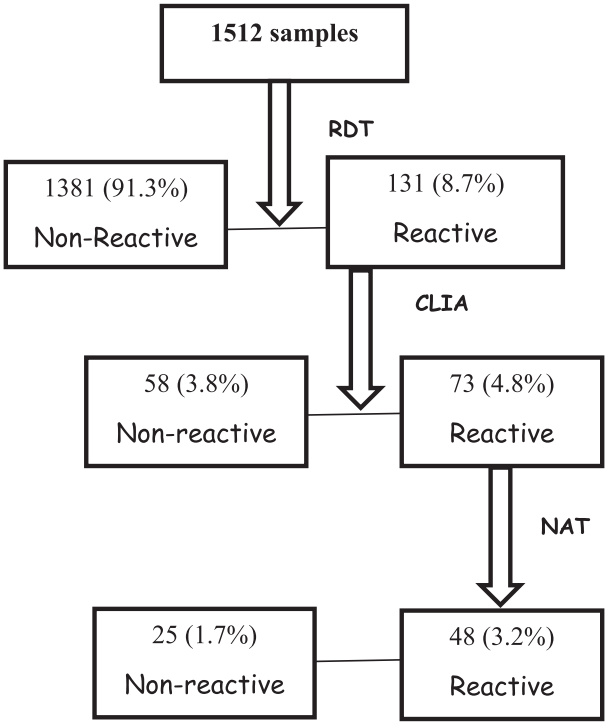


Fig. 1. The diagramm on HCV testing among blood donors in Lubumbashi.

4. Discussion

HCV infection is a major public health concern in transfusion, especially in countries with limited resources. Unlike HBV, which is transmitted not only by blood but also by sexual and maternal-fetal means, HCV is transmitted mainly by blood. HCV can also be transmitted through medical examinations and treatments (endoscopy, dental extraction, dialysis). We carried out this study to determine the seroprevalence, the molecular characterisation of HCV and the possible biochemical disturbances related to the carriage of anti-HCV antibodies in blood donors in Lubumbashi, DRC.

In this study, HCV seroprevalence was 4.8%. Only reactive samples in RDT and confirmed in CLIA were considered as positive samples and used in the determination of this seroprevalence. When considering the 131 blood donors with anti-HCV reactive in RDT, there was HCV seroprevalence of 8.7%. Based on previous HCV seroprevalence rates obtained using RDTs, this seroprevalence rate was consistent with those observed by Chukwurah et al [14] in Nigeria (7.6%) and Kabinda et al [13] in Bukavu (10%). However, it was significantly higher than those reported in other studies, notably those of Kabamba et al [15] in Kamina (0.2 %), Kakisingi et al [16] in Lubumbashi (2.67%), Kyalubile et al [17] in Kolwezi (0.7%), Lupande et al [6] in Kisantu (2.93%), and Mukendi et al [5] in Mbujimayi (1.9%). It should be noted that the seroprevalence of HCV Ab reported in the study of Kabinda et al [13] in Bukavu (10%) was in an HIV positive population. The observed differences in the HCV seroprevalence rates might be attributed to several factors including the geographic distribution of blood donors, risk factors, the quality of the selection of blood donors and the performance of the rapid screening tests for blood-borne microorganisms [18].

In the DRC, the screening for infectious markers in blood banks is carried out according to the recommendations of the national blood transfusion program (PNTS) and the national AIDS control program (PNLS) [6]. In the case of the hepatitis C virus, these programs recommend the use of rapid diagnostic tests (RDTs), for example "*HCV One Step Hepatitis C Virus Test Strip*" for the detection of anti-HCV antibodies. Although in developing countries many advantages have been attributed to these tests, it was demonstrated in a study carried out in Lubumbashi on the analytical performance of RDTs used in the screening and diagnosis of HCV and HBV that these tests presented a good sensitivity (100%) but their specificity was low (64%), particularly in the detection of anti-HCV antibodies, resulting in high rates of false positives [19]. The same study highlighted the need to submit any reactive sample in RDTs to a confirmation test that can be performed using a reliable method such as Enzyme Linked Immunosorbent Assay (ELISA), or Chemiluminescent Immunoassay (CLIA). ELISA and CLIA are extremely sensitive and specific techniques for the detection of anti-HCV antibodies in people at risk [20,21].

In fact, within the framework of this study, we relied on the analytical performance of these two techniques (RDT and CLIA) for screening of HCV among blood donors in Lubumbashi. RDTs allowed the detection of anti-HCV antibodies while CLIA confirmed the specificity of reactive antibodies in RDT. NAT (TMA) was used to determine the rate of chronic hepatitis C among the confirmed

Table 1
Seroprevalence of HCV in blood donors according to their socio-demographic characteristics.

Variable	n = 1512	Anti-HCV +		
General seroprevalence	4.8% (2.84 – 4.84)			
Gender		N (%)	OR (IC95%)	p
Male	1081	57 (5.3%)	2.25 (1.09 – 3.31)	0.03
Female	431	16 (3.7%)	0.16 (0.16 – 1.08)	
Age (years)				
[18 –29]	389	15 (3.9%)	1.2 (0.5 – 1.71)	0.17
[30–44]	675	37 (5.5%)	2.6 (1.8 – 4.9)	
[45–65]	448	21 (4.7%)	1.4 (0.8 – 2.1)	
Civil status				
Married	721	30 (4.2%)	1.7 (0.9 – 2.7)	0.11
Single	764	42 (5.5%)	2.4 (0.7 – 4.6)	
Divorced	27	1 (3.7%)	1.3(1.1 – 3.1)	
Occupation				
Liberal profession	545	28 (5.1%)	1.4 (0.4 – 4.3)	0.21
Government employees	528	22 (4.2%)	1.2 (0.3 – 1.7)	
Students	314	17 (5.4%)	1.6 (0.6 – 3.8)	
Unemployed	125	6 (4.8%)	1.2 (0.4 – 5.3)	
Categories of blood donors				
Regular volunteers	108	1 (1%)	1	0.02
Irregular volunteers	286	12 (4.2%)	1.2 (0.5 – 1.9)	
Family	1118	60 (5.4%)	3.9 (1.5 – 5.63)	

Table 2
Biochemical parameters in HCV positive blood donors versus HCV negative blood donors.

Paramètres	**Anti-VHC** + (n = 33)	**Anti-VHC - (n = 33)**	OR (IC95%)	*p*
CL T (<200 mg/dL)	207.4 ± 3.2	174.2 ± 5.8	1.2 (0.9 – 2.37)	0.09
HDL-C (>45 mg/dL)	39.3 ± 4.1	45.2 ± 3.7	1.8 (1.1 – 2.22)	0.14
LDL-C(<130 mg/dL)	144.3 ± 8.7	141.2 ± 9.4	0.9 (0.5 – 2.74)	0.71
TG (<130 mg/dL)	143.8 ± 8.2	135.2 ± 2.4	2,7 (1.1 –3.91)	0.21
Bil T (<12 mg/L)	15.7 ± 3.2	12.7 ± 2.8	4.1 (1.7 – 5.94)	0.02
Bil D (<2 mg/L)	4.7 ± 2.6	1.7 ± 1.4	3.3 (1.8 – 3.99)	0.01
Bil I (<10 mg/L)	12.6 ± 3.3	11.2 ± 2.8	1.7 (0.7 – 2.77)	0.12
ALAT (8 – 35 UI/L)	71.2 ± 3.8	19.7 ± 6.3	3.7 (1.5 – 5.11)	0.01
ASAT (8 – 30 UI/L)	54.3 ± 4.9	24.8 ± 4.7	3.1 (1.3 – 3.92)	0.01
PAL (30 – 100 UI/L)	98.4 ± 6.4	48.4 ± 7.5	2.4 (1.02 – 3.82)	0.03
γ GT (15 – 60 UI/L)	54.8 ± 4.5	39.8 ± 6.1	1.6 (0.7 – 2.98)	0.06
LDH (190 – 400 UI/L)	233.2 ± 4.6	221.5 ± 6.5	1.6 (0.19 – 2.16)	0.27
Alb (35 – 50 g/L)	37.9 ± 6.7	39.6 ± 3.6	1.9 (0.8 – 3.31)	0.17
CRP (<6 mg/L)	6.3 ± 1.7	4.6 ± 2.9	1.1 (0.47 – 2.64)	0.09

OR: odds ratio; IC: interval confidence; p: probability.

Table 3
Mutations and resistance to antiviral drugs.

G	n (%)	ST	n = 20	CV	M	MM	Antiviral drugs		Antiviral drugs		Antiviral drugs	
							NS5A		NS5B		NS3	
				log IU/mL	N (%)	Type	R	S	R	S	R	S
3	1(5.0)	3a	1 (5.0)	3.5 ± 2.1	-		None	DCV, ELB, LDV, OBV, PIB, VEL, SOF, GLE, GZR, PTV, SMV, VOX	None	DCV, ELB, LDV, OBV, PIB, VEL, SOF, GLE, GZR, PTV, SMV, VOX	None	DCV, ELB, LDV, OBV, PIB, VEL, SOF, GLE, GZR, PTV, SMV, VOX
4		4c	9 (45.0%)	4.2 ± 2.3	2 (22.2)	30R, 58P	DCV	ELB, LDV, OBV, PIB, VEL, SOF, GLE, GZR, PTV, SMV, VOX	None	DCV, ELB, LDV, OBV, PIB, VEL, SOF, GLE, GZR, PTV, SMV, VOX	None	DCV, ELB, LDV, OBV, PIB, VEL, SOF, GLE, GZR, PTV, SMV, VOX
	18 (90.0)											
		4 k	6 (30.0%)	3.9 ± 1.8	-	-	None	DCV, ELB, LDV, OBV, PIB, VEL, SOF, GLE, GZR, PTV, SMV, VOX	None	DCV, ELB, LDV, OBV, PIB, VEL, SOF, GLE, GZR, PTV, SMV, VOX	None	DCV, ELB, LDV, OBV, PIB, VEL, SOF, GLE, GZR, PTV, SMV, VOX
		4r	3 (15.0%)	3.6 ± 1.3	-	-	None	DCV, ELB, LDV, OBV, PIB, VEL, SOF, GLE, GZR, PTV, SMV, VOX	None	DCV, ELB, LDV, OBV, PIB, VEL, SOF, GLE, GZR, PTV, SMV, VOX	None	DCV, ELB, LDV, OBV, PIB, VEL, SOF, GLE, GZR, PTV, SMV, VOX
7	1 (5.0)	-	1 (5.0%)	5.9 ± 1.7	-	-	None	DCV, ELB, LDV, OBV, PIB, VEL, SOF, GLE, GZR, PTV, SMV, VOX	None	DCV, ELB, LDV, OBV, PIB, VEL, SOF, GLE, GZR, PTV, SMV, VOX	None	DCV, ELB, LDV, OBV, PIB, VEL, SOF, GLE, GZR, PTV, SMV, VOX
T	20 (100)		20 (100%)	M = 4.2 ± 1.8	2 (10.0%)							

G: genotype, T: total, ST:sub-genotypes, VL: viral load, M: mean, M: mutants, MM: marked mutations, N: number, R: resistance, S: sensitvity, DCV: daclatasvir, LDV: ledipasvir, ELB: elbasvir, OBV: ombitasvir, PIB: pibrentasvir, VEL: velpatasvir, SOF: sofosbuvir, GLE: glecaprevir, GZR: grazoprevir, PTV: paritaprevir, SMV: simeprevir, VOX: voxilaprevir.

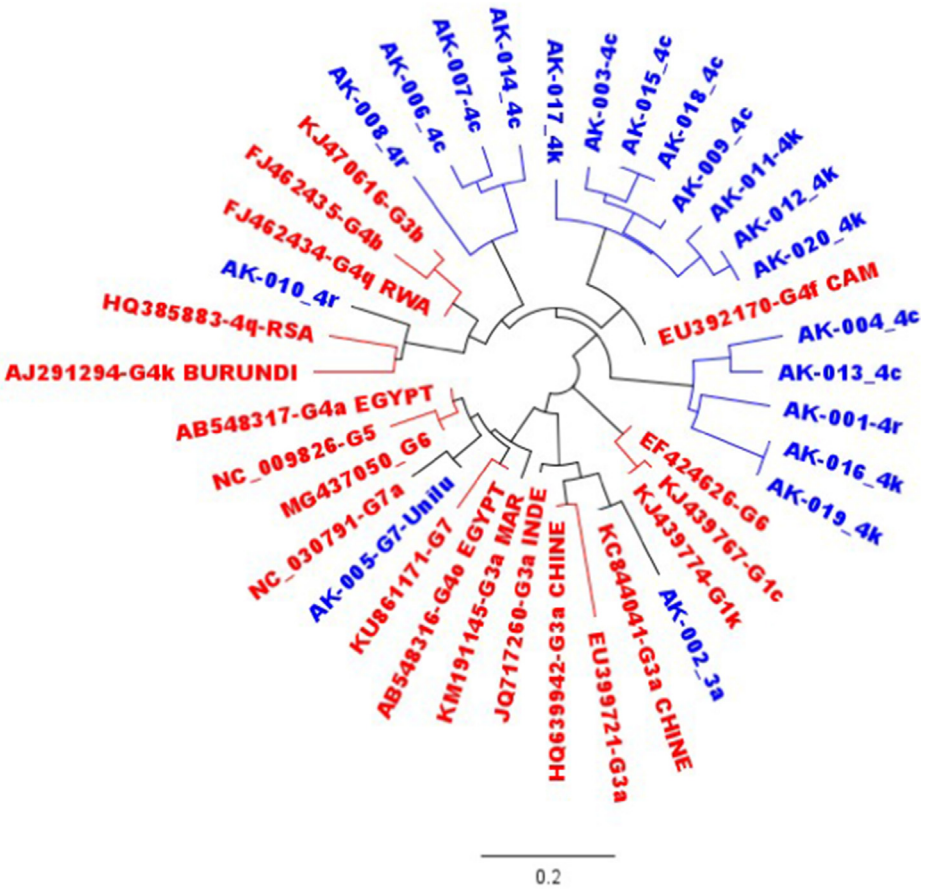


Fig. 2. Phylogenetic tree of representative HCV genotypes and sub-genotypes of Congolese blood donors in Lubumbashi. This tree derives from an alignment of a set of 20 HCV RNA sequences from Lubumbashi blood donors. These sequences relate to posterior probabilities > 0.60. The sequences in this study begin with the initials "AK" and are colored blue, while the reference sequences begin with other letters and are colored red.

seropositive cases, and to determine viral load for sequencing and genotyping. After satisfying the pre-selection conditions as mentioned in the pre-screening and clinical screening phases, 1512 samples were tested in RDT for the detection of anti-HCV antibodies, of which 131 (8.7%) were reactive. Reactive samples were tested in CLIA and 73 samples were found to be positive (4.8%). The interest of RDTs no longer needs to be demonstrated, particularly in developing countries. They are cheap, easy to use, do not require qualified personnel, speed of results and are of good conservation [22]. In order to estimate the rate of chronic hepatitis C among seropositive cases, the 73 reactive samples in CLIA were submitted to NAT and 48 (3.2%) of them have been found to be positive. NAT are heavy and expensive tests, requiring infrastructures adapted to their use and qualified personnel. In our study, CLIA reduced the initial seroprevalence by half (8.7% to 4.8%) and the NAT reduced it by third (8.7% to 3.2%). Without this confirmation test, about 58 non-reactive blood donor candidates in CLIA and 25 non-reactive in NAT would have had to be excluded from blood donating. Confirmatory tests are then important in estimating the seroprevalence of anti-HCV antibodies since they reduce the rate of false positives [23]. The reasons that underlie false positives in HCV screening are not yet clearly established. Parry et al [24] showed in their study that the causes of false reactivity can be diverse. These could be cross-reactions caused by immunological reactions to other pathogens, immunizations and autoantibodies. On the other hand, it seems that the poor analytical performance of methods for detecting infectious markers in the blood are associated with shortcomings in laboratory procedures such as calibration anomalies, contamination, storage conditions, etc. [25]. When considering the 73 blood donors anti-HCV reactive in CLIA, there was a concordance between CLIA and NAT in 48 cases (66%). In a study carried out in Brazil, on the "profile of blood donors in relation to hepatitis C" [21], there was a concordance between the methodologies electrochemiluminescence and NAT in 31 cases (37%) and disagreement in 53 cases (63%). The results of our study are different from those of Fröhlich since the two studies were not carried out in the same conditions, including geographical location of blood donors (DRC vs Brazil) and methodology (Chemiluminescent immuno-assay vs Electrochemiluminescent assay). Furthermore, in Fröhlich's study, all samples were tested both in serology and in NAT while in our study, samples were first tested in RDT, then reactive samples were tested in CLIA. Finally, only reactive samples in CLIA were tested in NAT. Similarly, in a study carried out on "high rate of HCV and HIV false-positive results in serologic screening in sub-Saharan Africa" [23], blood donations were first tested for HCV with both RDTs and EIAs. Reactive samples were confirmed with antigen/antibody assays, immunoblots and NAT. The HCV reactivity rate with RDTs was lower than with EIAs (0.55% vs. 3.52%). These results were not in agreement with those of this study (8.7% vs. 4.8%) in which CLIA was used as confirmatory test instead of EIA. However, there was a concordance between serologic screening and NAT of 58%. These results agreed with those obtained in this study (66%) although the two studies were not performed in the same conditions.

Sociodemographic characteristics such as gender and category of blood donors were significantly associated with the risk of

HCV infection. Indeed, in the context of our study, the rate of contamination was significantly higher among men than among women. Similarly, family donors had higher Anti-HCV Ab rate than irregular voluntary donors. These results agreed with those of the studies conducted by Kyalubile et al [17] and Mukendi et al [5]. Indeed, in Sub-Saharan Africa, there are more family donors than other categories of donors. Moreover, according to certain convictions, African society considers that blood donation from a family member would be of better quality than that of any other donor. In some emergency situations, family donations would not be tested prior to transfusion in some areas. Therefore, in developing countries, family blood donors are considered as an important risk factor for HCV infection.

Disturbance of biochemical parameters had been observed in positive anti-HCV antibodies blood donors. This was particularly the case for total and direct bilirubin, GPT, GOT and PAL. Our results corroborated those of Gronbaek et al [26] who had found in their study carried out in Denmark an increase in transaminases in positive anti-HCV antibodies blood donors. Since viral hepatitis C is a pathology affecting the liver, it is common to observe a disturbance of these biochemical parameters, reflecting the level of damage to hepatocytes. The mechanism underlying liver damage during HCV infection is not fully known, but it could be due to a combination of two mechanisms: firstly, the unlikely direct cytopathic effect of HCV on hepatocytes and, secondly, the well-known immune-mediated cell damage triggered by HCV [27]. Our results were also in agreement with those reported by Fahim et al [28] who observed in their study on biochemical changes in patients infected with schistosomiasis and hepatitis C a significant increase in direct and indirect bilirubin accompanied by a significant reduction in total protein and albumin. It should be important to remind that albumin is the most abundant plasma protein and is also used to transport bilirubin to the liver. Its synthesis is essentially hepatic, and its plasma level decreases in liver disease. Unlike non-alcoholic fatty liver disease, which is generally associated with hyperlipidemia, chronic hepatitis C is strongly linked to hypolipidemia, including hypocholesterolemia and hypo-triglyceridemia [29]. This would justify the normal lipid levels in our positive anti-HCV antibodies blood donors.

Of the 73 samples from HCV positive blood donors, 20 samples with a viral load > 500 IU/mL were selected for sequencing analyses. Genotypes 3a (5.0%), 4 (90.0%) and 7 (5.0%) were identified on the 20 studied HCV sequences. Phylogenetic analysis had shown that the HCV genotype 3a strain was close to that found in China. This result did not surprise us, since HCV genotype 3 predominates in Asia (China, India, Japan) [30] and there are significant population movements between China, India, and the DRC. The identified HCV strains of genotype 4 belonged to sub-types 4c (50.0%), 4 k (33.3%) and 4r (16.7%). These genotypes are described in this study for the first time in Lubumbashi. Our results agreed with those of Iles et al [31] and Mongo-Onkouo et al [32] who reported a high prevalence of HCV genotype 4 respectively in Congo-Brazzaville and Kinshasa (DRC). It should be noted that infection with HCV genotype 4 is very common in Africa and particularly in Egypt where the proportion of subjects infected with HCV genotype 4 is the highest in the world [33]. Phylogenetic analyzes had shown that genotype 4 originated in central Africa and that several subtypes were later exported to other parts of Africa since the 19th century, including sub-type 4a which predominates today in Egypt [33,34,35,36]. The identified HCV sub-types of genotype 4 in this study were mostly close to those found in Cameroon, an African country located not far from the DRC. Although HCV genotype 4 has the reputation of being a poor responder to antiviral treatments, the current treatment of patients infected with HCV genotype 4 is based on the administration of the combination of Sofosbuvir and another molecule for a period of at least 12 weeks. The determination of the HCV genotype is imperative for the initiation and evaluation of the duration of treatment [37]. Regarding the genotype 7, its identification was made for the first time in this study in a blood donor residing in the DRC. Phylogenetic analysis had shown that the identified HCV genotype 7 in this study was close to that found in Uganda in two Congolese citizen from the DRC. It should be noted that HCV genotype 7 was identified for the first time in 2007 in Canada in an individual from the DRC [38].

The mean HCV viral load in infected blood donors was 4.2 ± 1.8 log IU/mL. HCV genotype 7 had the highest viral load (5.9 log IU/mL). Two of the 20 identified HCV genotype 4c positive blood donors had 30R and 58P mutations, responsible for resistance to daclatasvir (DCV). These mutations had not resulted in any resistance to the other direct antivirals.

The current study had a limitation related to the study population. Since the study population consisted only of blood donors, this study might not evaluate the seroprevalence of the HCV infection in the general population. Therefore, a large representative sample of the population is needed to study the exact distribution of HCV genotypes in Lubumbashi.

5. Conclusion

Our study focused on HCV infection seroprevalence and RNA-HCV genotypes distribution among blood donors in Lubumbashi. At the end of this study, the seroprevalence rate was 4.8% for anti-HCV antibodies in blood donors. Compared to the prevalence rates reported in previous studies among blood donors, we find that hepatitis C virus infections in Lubumbashi were on the rise. These results highlight the need to apply strategies in Lubumbashi aiming at strengthening blood safety, including a supply based on voluntary, regular and unpaid blood donations, effective education of the public on blood donation, the selection of blood donors and screening using high sensitivity tests, the use of concentrated red blood cells instead of whole blood and the introduction of techniques for inactivating any pathogens present in labile blood derivatives. This study identified for the first time in Lubumbashi HCV strains of genotypes 3a, 4 (sub-genotypes 4c, 4 k and 4r) and 7. Similarly, it identified some drug resistance mutations to antiviral drugs. These results could not only help in the implementation of viral hepatitis C treatment strategies but also contribute to the development of the mapping of HCV genotypes in Lubumbashi and in the DRC.

Finally, this study demonstrated that viral hepatitis C results in some disturbances of biochemical parameters that could contribute to the evaluation of the degree of liver function alteration.

Author contributions

All authors have read and approved the final version of this manuscript.

Conflict of interest

No conflict of interest was declared.

References

[1] Uwingabiye J, Zahid H, Unyendje L, Hadef R. Séroprévalence des marqueurs viraux sur les dons du sang au Centre de Transfusion sanguine, Hôpital Militaire d'Instruction Mohammed V de Rabat. Pan Afr Med J 2016;25:185.
[2] Bigna JJR, Amougou MA, Asangbeh SL, Kenne AM, Nansseu JR. Séroprévalence de l'infection par le virus de l'hépatite C au Cameroun: revue Systématique avec méta-analyse. BMJ Open 2017;7:28.
[3] Zeba MTA, Sanou M, Bisseye C, Kiba A, Nagalo BM, et al. Characterisation of hepatitis C virus genotype among blood donors at the regional blood

transfusion centre of Ouagadougou. Burkina Faso Blood Transfus 2012. https://doi.org/10.2450/2012.0089-12.
[4] Martinez-Rodriguez ML, Urebe-Noguez LA, Arroyo-Anduiza CI, Mata-Marin JA, Benitez-Arvizu G, et al. Prevalence and risk factors of occult hepatitis C infections in blood donors from Mexico City. PLoS One 2018;13: e0205659.
[5] Mukendi ANK, Ciamala MP, Kanyiki KM, Banza NDB, Mukendi MJR, et al. Seroprevalence of hepatitis C virus infection among blood donors in General Dipumba Hospital in Mbuji Mayi, Democratic Republic of Congo. J Hepatol Gallblader Dis Research 2018;2:1–4.
[6] Lupande MD, Busa MG, Pululu C, Mukuku O, Phoba MF, et al. Risk factors of the transmission of HIV, hepatitis B and C and Syphilis among blood donors at the saint Luc General Hospital of Kisantu, Democratic Republic of Congo. Journal of HIV & Retro virus 2018;4:18.
[7] Kchouk FH, Gorgi Y, Bouslama L, Sfar I, Ayari R, et al. Phylogenetic Analysis of Isolated HCV strains from Tunisian Hemodialysis Patients. Viral Immunol 2013;26(1):40–8. https://doi.org/10.1089/vim.2012.0043.
[8] Kabinda JM, Kembo Nsayi L, Kalala Kabange JN. Estimation des risques résiduels de transmission du VIH, des virus des hépatites B et C par la transfusion sanguine à Lubumbashi, République Démocratique du Congo (RD Congo). Journal Home 2019;21:12–7.
[9] Temesgen AB, Andamlak DE. Seroepidemiological patterns and predictors of hepatitis B, C and HIV viruses among pregnant women attending antenatal care clinic of Atat Hospital. Southern Ethiopia SAGE Open Medicine 2020;8:1–9. https://doi.org/10.1177/20503/2/11990870.
[10] Nii-Trebi NI, Brouwn CA, Osei YD, Ampofo WK, Nyarko AK. Core encoding Sequences of Hepatitis C Virus in Ghanaian blood donors are predominantly mosaics of different genotype 2 Strains and cannot distinguish subtypes. BMC Infect Dis 2019;19:533. https://doi.org/10.1186/s12879-019-4155-4.
[11] Navaneethan U, Kemmer N, Neff GW. Predicting the probable outcome of treatment in HCV Patients. Therap Adv Gastroenterol 2009;2:287–302. https://doi.org/10.1177/1756283X09339079.
[12] Kabamba AT, Kalunga BT, Mwamba CM, Nyembo CM, Dufrasne F, et al. Epidemiological aspects and molecular characterization of the hepatitis B virus among blood donors in Lubumbashi, Democratic Republic of Congo. Transfus Clin Biol 2021;28:30–7. https://doi.org/10.1016/j.tracli.2020.10.012.
[13] Kabinda JM, Katchunga P. Les hépatites virales B et C chez les porteurs du virus de l'immunodéficience humaine à Bukavu (sud-kivu), République Démocratique du Congo. J Afr Hépatol Gastroentérol 2010;4:230–5.
[14] Chukwurah EF, Ogbodo SO, Obi GO. Serprevalence of Hepatitis C Virus (HCV) infection among blood donors in a South-Eastern State of Nigeria. Biochemical Rearch 2005;16:133–5.
[15] Kabamba MN, Ilunga KB. Prévalence des marqueurs infectieux chez les donneurs de sang en milieu rural. Cas de l'hôpital général de référence de Kamina. Sante Publique 2013;25:213–7.
[16] Kakisingi NC, Mukuku O, Matanda KS, Manika MM, Kyabu KV, et al. Profil épidémiologique et séroprévalence des donneurs de sang aux cliniques universitaires de Lubumbashi, République Démocratique du Congo. Pan African Medical journal 2016;23:175. https://doi.org/10.11604/pamj.2016.23.175.8480.
[17] Kyalubile MNFS, Tshibanda CN, Tshivwadi AT, Kabwe DN, Sonnhy TT, et al. Seroprevalence of hepatitis B and C in the blood donors in Kolwezi, Democratic Republic of Congo. J Hepatol Gallblader Dis Research 2018;2:1–4.
[18] World Health Organization, WHO. Hepatitis B. Available at: htpp://www.who.int/mediacentre/factsheets/fs204/en/. Accessed Dec 1, 2016.
[19] Kabamba AT, Mwamba CM, Dessilly G, Dufrasne F, Kabamba BM, et al. Evaluation of the analytical performance of six rapid diagnostic tests for the detection of viral hepatitis B and C in Lubumbashi, Democratic Republic of Congo. J Virol Methods 2020 Nov;285:113961. https://doi.org/10.1016/j.jviromet.2020.113961.
[20] Fourati S, Feld JJ, Chevaliez S, Luhamann N. Approaches for simplified HCV diagnostic algorithms. J Int A'IDS Soc 2018;21.
[21] Fröhlich AKS, Brum D, Dallegrave E. Profile of blood donors in relation to hepatitis C. Int J Hematol Blo Dis 2018;3:1–7.
[22] D'Acremont V, Greub G, Genton B. Tests diagnostiques rapides (TDR): la panacée pour le praticien ? Rev Med Suisse 2011;7:984–90.
[23] Candotti D, Sauvage V, Cappy P, Boullahi MA, Bizimana P, et al. High rate of hepatitis C virus and human immunodeficiency virus false-positive results in serologic screening in sub-Saharan Africa: adverse impact on the blood supply. Transfusion 2020;60:106–16. https://doi.org/10.1111/trf.15593.
[24] Parry JV, Easterbrook P, Sands AR. One or two serological assay testing strategy, for diagnosis of HBV and HCV infection? The use of predictive modelling. BMC l'infect Dis 2017;17:705.
[25] Twagirumugabe T, Swaibu G, Bergström T, Walker TD, Gahutu JB, et al. Low prevalence of hepatitis C virus RNA in blood donors with anti-hepatitis C virus reactivity in Rwanda. Transfusion 2017 Oct;57:2420–32. https://doi.org/10.1111/trf.14204.
[26] Gronbaek KE, Jensen OJ, Krarup HB, Dietrichson O. Biochemical, virological and histopathological changes in Danish blood donors with antibodies to hepatitis C virus. Dan Med Bull 1996;43:186–8.
[27] Gerber MA, Thung SN. Immunopathology of hepatitis C. Immunology and liver falk symposium. Basel, Abstrack Book 1992;70:19–20.
[28] Fahim FA, Esmat AY, Hassan GK, Abdel-Bary A. Biochemical changes in patients with combined chronic schistosomiasis and viral hepatitis C infections. Dis Markers 2000;16:111–8.
[29] Chang ML. Metabolic alterations and hepatitis C: From bench to bedside. World J Gastroenterol 2016;22:1461–76.
[30] Tagajdid MR, Rahoui J, Elannaz H, El Kochri S. Les nouvelles techniques de génotypage du virus de l'hépatite C. Journal de Biologie Médicale 2014;3:106–8.
[31] Iles JC, Harrison GLA, Lyons S, Djoko CF, Tamoufe U, et al. Hepatitis C virus infections in the Democratic Republic of Congo exhibit a cohort effect. Infect Genet Evol 2013;19:386–94.
[32] Mongo-onkouo A, Ahoui Apendi CP, Mimiesse Monamou JF, Ontsira N, Ngalessami NM, et al. Distribution des Génotypes et Sous Types du Virus de l'Hépatite C au Congo-Brazzaville. Health Sci Dis 2018;19:33–6.
[33] Ray SC, Arthur RR, Carella A, Bukh J, Thomas DL. Genetic epidemiology of hepatitis C virus throughout egypt. J Infect Dis 2000;182:698–707.
[34] Angelico M, Renganathan E, Gandin C, Fathy M, Profili MC, et al. Chronic liver disease in the Alexandria governorate, Egypt: contribution of schistosomias and hepatitis virus infections. J Hepatol 1997;26:236–43. https://doi.org/10.1016/s0168-8278(97)80036-0.
[35] Franck C, Mohamed MK, Strickland GT, Lavanchy D, Arthur RR, et al. The role of parenteral antischistosomal therapy in the spread of hepatitis C virus in Egypt. Lancet 2000;355:56–8. https://doi.org/10.1016/s0140-6736(99)06527-7.
[36] Kamal SM, Madwar MA, Bianchi L, Tawil AE, Fawzy R, et al. Clinical, virological, and histological features: long-term follow-up in patients with chronic hepatitis C co-infected with S. mansoni. Liver 2000;20:281–9. https://doi.org/10.1034/j.1600-0676.2000.020004281.x.
[37] Cho JE, Jeong SH, Han BH, Lee SU, Yun BC, et al. Hepatitis C virus (HCV) genotypes and the influence of HCV subtype 1b on the progression of chronic hepatitis C in Korea: a single center experience. Clin Mol Hepatol 2012;18:219–24.
[38] Murphy DG, Sablon E, Chamberland J, Fournier E, Dandavino R, et al. Hepatitis C Virus Genotype 7, a New Genotype Originating from Central Africa. J Clin Microbiol 2015 Mar;53:967–72. https://doi.org/10.1128/JCM.02831-14.